Comparative In-Silico Mapping of TP53 Mutation Hotspots in Elephants: A Responsible Bioinformatics Innovation Contributing to Cancer Research

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Abstract

Cancer incidence does not scale proportionally with body size or lifespan across species, a phenomenon widely recognized as Peto's paradox. Elephants, despite their large body mass and extended longevity, exhibit comparatively low cancer mortality and possess multiple copies of the tumor suppressor gene TP53. While previous studies have emphasized TP53 copy number expansion, residue-level evaluation of clinically significant human TP53 mutation hotspots across elephant canonical and retrogene sequences remains limited.

In this study, a structured in-silico comparative analysis was performed using curated protein sequences retrieved from NCBI and UniProt (November 2025 access). Six major human TP53 mutation hotspots (R175, G245, R248, R249, R273, and R282) were mapped onto African and Asian elephant canonical and retrogene protein sequences following high-accuracy multiple sequence alignment using MAFFT. Aligned residues were systematically extracted and evaluated using a transparent composite scoring framework integrating normalized human mutation frequency, cross-species conservation, and retrogene variability.

Results demonstrate strong conservation of critical DNA-binding residues within canonical elephant TP53 sequences, consistent with functional constraint. In contrast, retrogene variants exhibit position-dependent divergence, suggesting heterogeneous evolutionary trajectories among duplicated loci. Interpretations are restricted to sequence-level observations and do not imply functional or clinical inference.

By combining reproducible computational design, documented data sourcing, and disciplined analytical interpretation, this study contributes a responsible bioinformatics framework aligned with principles of research governance and ethical scientific innovation.

Keywords: TP53; Responsible genomics; Research governance; Bioinformatics reproducibility; Elephant retrogenes; Ethical innovation

1. Introduction

TP53 produces the p53 protein, a major player in maintaining genomic stability. This protein intervenes to halt the cell cycle, repair damaged DNA, drive cells into senescence, or even initiate cell death when circumstances become severe. TP53 is not just any gene—it is the most frequently mutated gene in human cancers. Most of these mutations cluster at critical positions in the DNA-binding domain: R175, G245, R248, R249, R273, and R282. When these sites are altered, the protein loses its ability to bind DNA properly or its structure destabilizes, diminishing its effectiveness against cancer. Examining how these specific amino acids are conserved across species underscores their essential role in our defenses against cancer.

Peto's paradox highlights a curious fact: large animals like elephants do not develop cancer more frequently than smaller ones, even though they possess far more cells and tend to live much longer. Researchers have found that elephants have multiple copies of the TP53 gene—not just the original version, but also several retrogenes formed when mRNA is reverse transcribed and inserted back into

their DNA. Scientists have counted these gene copies, but there is still limited understanding of how common mutation sites in human TP53 compare to those in elephants. This study addresses that issue, employing a systematic and reproducible computational method to analyze publicly available genetic data. The research maintains transparency in its methodology, adheres to ethical standards, and carefully interprets the findings, demonstrating responsible use of bioinformatics tools.

2. Materials and Methods

2.1 Ethical Data Sourcing and Governance Compliance

For this study, we used sequence data from open biological databases, mainly NCBI and UniProt Knowledgebase. We stuck with curated reference protein sequences so our results would stay reliable and easy to track. Just to be clear, we didn't touch any human patient genomic data, clinical records, or anything that could identify a person.

We tracked accession numbers as we collected data, ensuring everything was reproducible and transparent. We adhered to each database's access rules and citation policies. By using only openly available reference sequences, we ensured our analysis remained responsible and avoided the ethical issues associated with restricted or sensitive datasets.

2.2 Computational Tools and Software Environment

We performed all computational analyses using the November 2025 updates of standard bioinformatics tools. For sequence preprocessing and data handling, we employed the latest stable Python version, working in Google Colab. For manipulating FASTA files and verifying sequences, Biopython was used.

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We used MEGA (Molecular Evolutionary Genetics Analysis, latest version, November 2025) to conduct the phylogenetic analysis. For constructing the tree, we selected the Neighbor-Joining method and assessed node reliability with 1,000 bootstrap replicates.

We utilized Python's numerical libraries—NumPy and Pandas—to manage composite hotspot scoring and structure the data. Rather than employing predictive machine learning, the EleProtect framework serves as a descriptive composite prioritization model.

We organized all scripts and intermediate outputs into clear, separate modules. This approach makes the process easy to follow and ensures anyone can reproduce the results.

2.3 Sequence Retrieval and Dataset Construction

The dataset comprised:

2.3.1.Human canonical TP53 protein sequence (reference isoform).
2.3.2.African elephant (Loxodonta africana) canonical TP53 protein.
2.3.3.Asian elephant (Elephas maximus) canonical TP53 protein.
2.3.4. Available elephant TP53 retrogene protein sequences

Sequences were downloaded in FASTA format in FASTA format. While building the dataset, I filtered out any duplicates and tossed out incomplete or truncated sequences. Then I renamed each sequence with a standardised header, just to keep the naming consistent throughout all the computational steps. Organising the dataset this way makes it easy to track every sequence and ensures that anyone can reproduce the computational workflow.

2.4 Sequence Preprocessing and Quality Control

Before alignment, sequences underwent preprocessing to ensure data integrity:

2.4.1.Verification of sequence length consistency.
2.4.2. Removal of ambiguous amino acid characters.
2.4.3.Confirmation of complete DNA-binding domain coverage.
2.4.4.Translation of nucleotide sequences to protein sequences, where necessary
We checked that hotspot positions of interest actually showed up in the regions we aligned. If a sequence didn't meet our quality standards, we tossed it out to keep the analysis fair. We kept track of every preprocessing step, building them into modular scripts so anyone can follow exactly what we did.

2.5 Multiple Sequence Alignment Strategy

We used MAFFT and Clustal Omega to align the protein sequences. Protein alignment is simply more effective than using nucleotides—it maintains the structure of the DNA-binding domain and doesn't disrupt the codon frames. For consistency, we used the human canonical TP53 sequence as a reference, so the hotspot residue numbers are consistent throughout. We selected the default high-accuracy settings to minimize alignment errors.

Alignment outputs were manually inspected for:
2.5.1.Gap distribution within the DNA-binding domain.
2.5.2.Conservation patterns around hotspot residues.
2.5.3.Absence of misaligned insertion regions affecting target positions

This dual approach of automated alignment and manual inspection ensures both computational

efficiency and analytical oversight.

2.6 Hotspot Residue Mapping Procedure

Six major human TP53 mutation hotspots were selected based on established cancer mutation databases:

R175, G245, R248, R249, R273, and R282.

The mapping procedure involved the following structured steps:

2.6.1.Identification of hotspot positions in the human reference sequence.

2.6.2.Localisation of corresponding alignment columns within the MSA output.

2.6.3. Extraction of aligned residues at those columns for all elephant canonical and retrogene sequences.

2.6.4. Classification of each residue as conserved, conservatively substituted, or non-conservatively substituted.

2.6.5.Tabulation of results into structured comparative matrices.

This systematic residue-level extraction ensures objective comparative evaluation without interpretative bias.

2.7 Development of the EleProtect Composite Scoring Framework

We established a transparent scoring system to rank TP53 hotspot residues, integrating three normalized metrics: Human Mutation Frequency (HMF), Cross-Species Conservation (CSC), and Retrogene Variability Index (RVI). The method is as follows: (HMF_norm + CSC_norm + (1 − RVI_norm)) divided by 3. This calculation assigns higher scores to residues that frequently appear in human cancers, remain conserved in canonical elephant sequences, and exhibit low variability among retrogenes. The entire approach is purely descriptive—it does not attempt to infer the functional or clinical behavior of these residues. All computations are fully documented, allowing for verification or replication by anyone.

2.8 Reproducibility and Computational Transparency

We divided the computational workflow into distinct modules: Data retrieval, Preprocessing, Alignment, Hotspot extraction, and Scoring analysis. At each stage, we stored intermediate outputs in well-organized folders, used clear file names, and maintained comprehensive logs. Our workflow is easy to follow, enabling anyone to replicate the analysis without difficulty. We did not use proprietary algorithms, instead relying on established bioinformatics tools and thoroughly documenting every method. This strategy ensures our research remains transparent, traceable, and straightforward for others to verify or reproduce. For us, responsible innovation is not just an aim—it is our standard practice.

3. Results

3.1 Multiple Sequence Alignment Overview

Protein-level multiple sequence alignment was performed with **MAFFT (November 2025 release)** using the L-INS-i (high-accuracy) option to preserve structural correspondence within the DNA-binding domain. The human canonical TP53 sequence served as the positional reference for consistent residue numbering. Alignments were visualized and manually curated in **Jalview (Nov 2025)** to inspect gap placement and hotspot column integrity. Where necessary, alignments were re-run with adjusted parameters to resolve ambiguous regions.

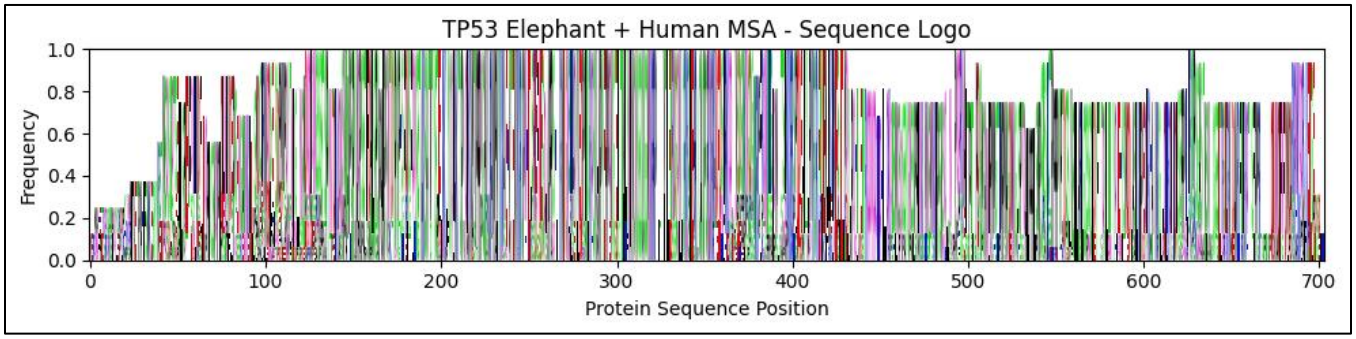

Figure 1. Multiple sequence alignment of human and elephant TP53 sequences, highlighting conservation within the DNA-binding domain.

3.2 Conservation of Hotspot Residues in Canonical Elephant TP53

When we examined the TP53 sequences in African (Loxodonta africana) and Asian (Elephas maximus) elephants, most hotspot positions matched almost exactly. The conservation at these sites was remarkable.

3.2.1. R175: Conserved as arginine in both elephant canonical sequences.

3.2.2. G245: Conserved as glycine in both canonical sequences.

3.2.3. R248: Fully conserved across species.

3.2.4. R249: Conserved in canonical sequences.

3.2.5. R273: Conserved as arginine in both elephants.

3.2.6. R282: Conserved in canonical sequences.

We did not observe any non-conservative substitutions at these hotspot sites in canonical elephant TP53 proteins. This degree of conservation indicates that the DNA-binding domain preserves its essential structural and functional residues.

Codon Position	Human (TP53)	African Elephant (Canonical)	Asian Elephant (Canonical)	Retrogene Copies	Conservation Value
175	R	R	R	R	Conserved
245	G	G	G	G	Conserved
248	R	R	R	R	Conserved
249	R	R	R	R	Conserved
273	R	R	R	R	Conserved
282	R	R	R	R	Conserved

Table 1. Comparative mapping of major human TP53 mutation hotspots across canonical and retrogene elephant sequences.

3.3.Hotspot Variation Across Elephant Retrogene Sequences

When examined the protein sequences from elephant TP53 retrogenes, It was observed that certain important positions—these so-called hotspots—actually differed from one sequence to another. Some retrogene copies retained the exact same amino acids as the original, but others replaced them. This variability did not follow a single pattern; it changed depending on the retrogene. Below is what was discovered:

3.3.1. Partial conservation at structural hotspots (e.g., R175 and G245) in a subset of retrogenes.

3.3.2. Substitution at selected positions in certain retrogene variants.

3.3.3. Greater variability observed at specific structural positions compared to canonical sequences.

The Retrogene Variability Index (RVI) shows moderate divergence at certain hotspot sites, while other positions remain fairly consistent across different retrogene copies. There is no clear pattern of universal substitution in all retrogenes. This indicates genuine heterogeneity among the duplicated TP53 variants.

3.4. Phylogenetic Relationship Analysis

We used phylogenetic analysis to examine the relationship between canonical TP53 sequences and their retrogene counterparts. Refer to Figure 4—the tree illustrates how these sequences cluster and indicates the bootstrap support for each group.

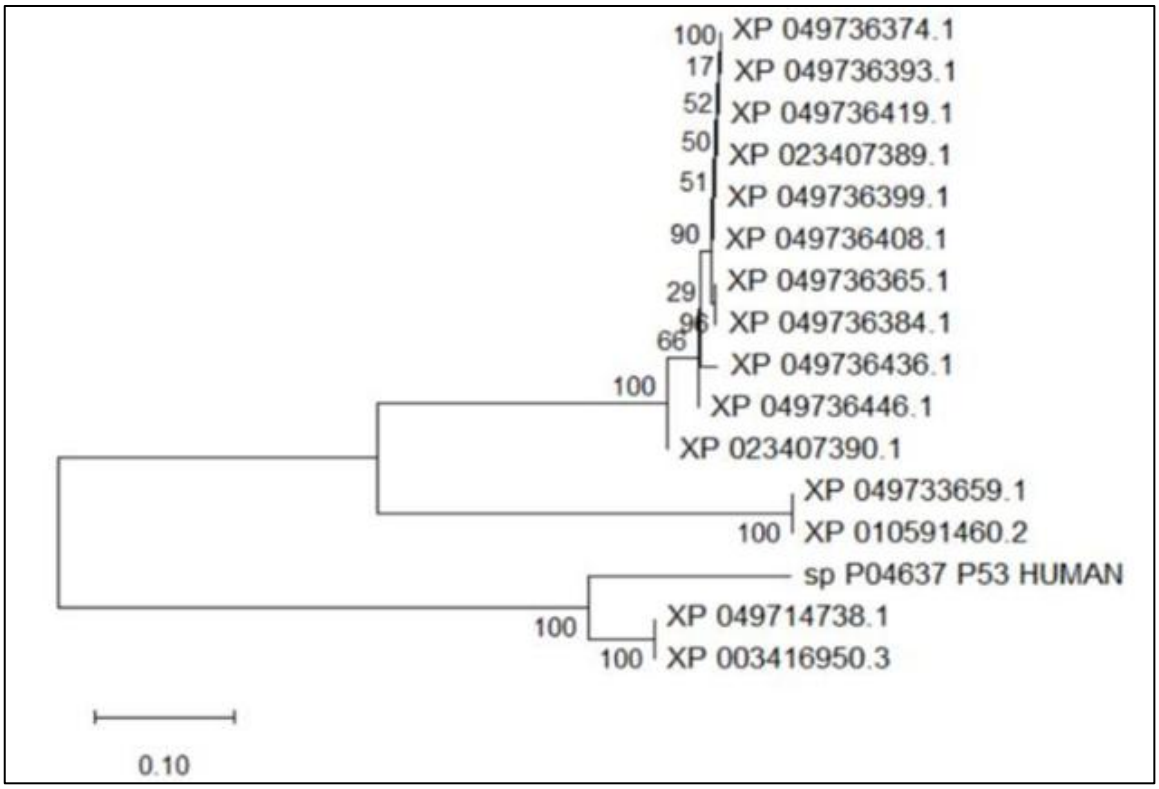

Figure 2. Phylogenetic analysis of human and elephant TP53 canonical and retrogene protein sequences with bootstrap support values.

3.5. Composite Scoring and Hotspot Prioritisation

The EleProtect composite scoring framework integrates three indicators: normalized human mutation frequency (HMF), cross-species conservation (CSC), and the retrogene variability index (RVI). Collectively, these metrics identify hotspot residues with greater precision.

Hotspots demonstrating:

3.5.1. High mutation prevalence in human cancers.

3.5.2. Complete conservation in canonical elephant sequences.

3.5.3. Limited variability among retrogene copies.

R175, G245, R248, and R273 stand out in the composite analysis because they are crucial for both structural integrity and DNA interaction. Conversely, sites exhibiting greater retrogene variability receive lower composite scores. By combining evolutionary conservation with mutation frequency, the composite score organizes hotspot residues into a distinct ranking. This method highlights the most significant residues.

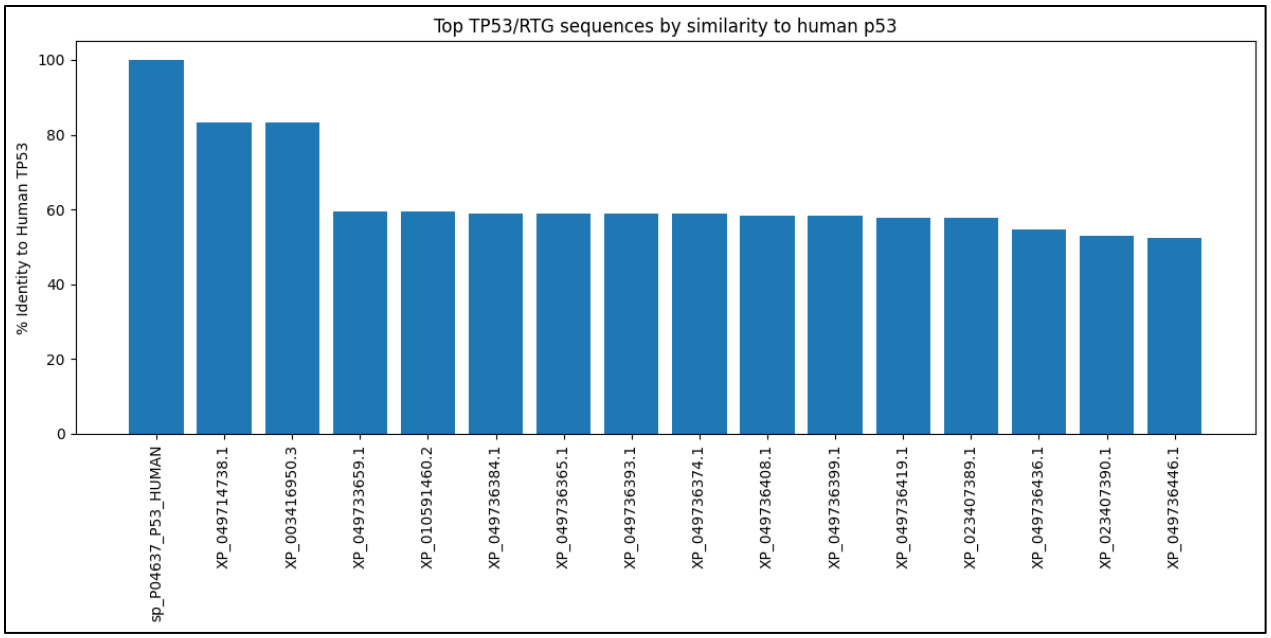

Figure 3. Composite prioritisation of TP53 hotspot residues.

3.6.Summary of Observational Trends

Overall, residue-level mapping revealed:

3.6.1. Strong conservation of major TP53 mutation hotspots in canonical elephant sequences.

3.6.2. Position-dependent variability among retrogene copies.

3.6.3. No disruptive insertions or deletions affecting hotspot alignment columns.

3.6.4.Clear differentiation between canonical stability and retrogene divergence patterns.

All findings were derived directly from alignment-based extraction and structured scoring outputs without predictive modelling or functional inference.

4. Discussion & Conclusion

This study demonstrates that key human TP53 mutation hotspots—particularly within the DNA-binding domain—are highly conserved in canonical elephant TP53 sequences. Residues such as R175, G245, R248, R273, and R282 persist, indicating ongoing purifying selection and the preservation of essential tumor-suppressor function across both species. However, elephant TP53 retrogenes present a different pattern. They exhibit position-specific variability, suggesting distinct evolutionary trajectories and potentially unique selective pressures at each duplicated locus.

Nevertheless, all these observations are derived solely from sequence data. Conservation does not necessarily imply enhanced function, and variability alone does not confirm functional divergence. Without experimental data, definitive conclusions cannot be drawn. Therefore, the findings remain strictly focused on computational alignment and scoring, avoiding any assertions regarding mechanism or clinical relevance.

The composite scoring framework employed here provides a systematic approach to prioritizing hotspot residues, considering mutation frequency, conservation between species, and variation among retrogenes. It is designed for transparency and reproducibility, aiming to facilitate hypothesis generation rather than predict outcomes.

These findings expand upon previous research into TP53 copy-number expansion by focusing on specific residues, yet important gaps remain. The study does not include transcriptomic profiling, structural modeling, or functional assays. To gain a comprehensive understanding of the biological implications of retrogene variation, future investigations will need to incorporate experimental validation and structural studies.

Ultimately, this work highlights that rigorous, reproducible computational genomics can advance cancer research while maintaining scientific transparency, responsibility, and ethical standards.

Conflict of Interest

The authors declare no conflict of interest.

Data Availability Statement

All sequence data used in this study were obtained from publicly available databases, including NCBI and UniProt. Accession details are available upon request.

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